

Alesio Puf



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Education

Babes-Bolyai University

Bachelors in Computer Science GPA: 9.42 / 10.00

Sep 2023 – Jul 2026

Cluj-Napoca, Romania

- **Top 5% of class** – Merit Scholarship (2023–2026)
- Collaborated on **EU Sustainable Development Goals** initiative by developing a **real-time disaster response platform**

Experience

dotLumen

R&D Software Engineer

Mar 2026 – present

Cluj-Napoca, Romania

- Developed a dynamic configuration bridge that interfaces **ML-based environment classification** with **Nav2 costmap parameters**, automating real-time adjustments for path-planning.
- Initiated the migration of the core navigation stack from **ROS to ROS2**, refactoring legacy nodes to utilize **managed lifecycles** and improved real-time execution for safety-critical operations.
- Integrated **Intel RealSense D455** sensors into the hardware abstraction layer, performing **extrinsic/intrinsic calibration** and firmware tuning to extend depth-sensing range by **2x** compared to the **D435**.

Amazon

Software Development Engineer Intern

Jul 2025 – Nov 2025

Madrid, Spain

- Scaled **Invoice by Amazon** services across regional central nodes to improve **latency** and **availability**, thus supporting a **5%** increase in transaction throughput during peak traffic.
- Improved workflow velocity for on-call and QA teams by **10%** by designing and implementing a streamlined solution, leveraging **Lambda, DynamoDB, SQS** and other **AWS Technologies** to automate complex manual procedures.

Thales

Backend Developer

Mar 2022 – Jun 2025

Remote

- Developed a microservice-based **passenger information system** web application deployed for urban and regional **train infrastructure** in key regions of the world (**Paris, Lisbon, Dubai, Cairo, Manila**).
- Engineered **driver-level microservices** for audio device integration, improving system functionality through low-level protocol implementations, resulting in **4 new major clients** onboarded.
- Initiated the migration from a legacy **Java 8** monolith to **Java 21** microservices, incorporating modern technologies for infrastructure and **CI/CD**, resulting in a **20%** reduction of cycle time.

Projects & Awards

Romanian HUB for Artificial Intelligence (HRIA) | TensorFlow, PyTorch, OpenCV, Hugging Face

Sep 2023 - present

- Completed analysis and normalization of **900+ data points** by implementing **automated preprocessing pipelines**, resulting in **3-week acceleration** of project timeline.
- Developed **7 new methodologies** for assessing suture quality in anastomotic lapse index analysis through implementation of convolutional networks and computer vision algorithms, achieving a **3.6% false negative rate** and **82% accuracy** for automated medical diagnoses.

Day Zero Platform | React, Tailwind CSS, Spring Boot, PostgreSQL, AWS

Nov 2024

- Led a cross-functional team of **5 developers** to design and implement a **cybersecurity platform** that transforms traditional **CTF challenges** into **PVP experiences**, securing **1st place** among **30+ teams** in the hackathon.

Raiffeisen Bank Think Tank | React Native, Bootstrap, Spring Boot, MySQL, AWS

Dec 2023 – Apr 2024

- Developed a secure **digital stock trading platform** for the **Bucharest Stock Exchange**, enabling **real-time trading**, **AI-based sentiment analysis**, and an **education portal**, winning **1st place** among **15+ teams**.

Technical Skills

Languages: Python, Java, C++, TypeScript, C, Rust

Technologies: ROS/ROS2, Nav2, Intel RealSense SDK, OpenCV, TensorFlow, PyTorch, AWS (Lambda, DynamoDB, SQS), Docker, Spring Boot, Jenkins, Redis, RabbitMQ, Kafka, SQL, React, Node.js, gRPC

Core Concepts: Robotics Middleware, Real-Time Systems, SLAM, Sensor Fusion, Computer Vision, Microservices Architecture, Event-Driven Systems, Artificial Intelligence, Machine Learning, Retrieval-Augmented Generation (RAG), Agile Methodologies